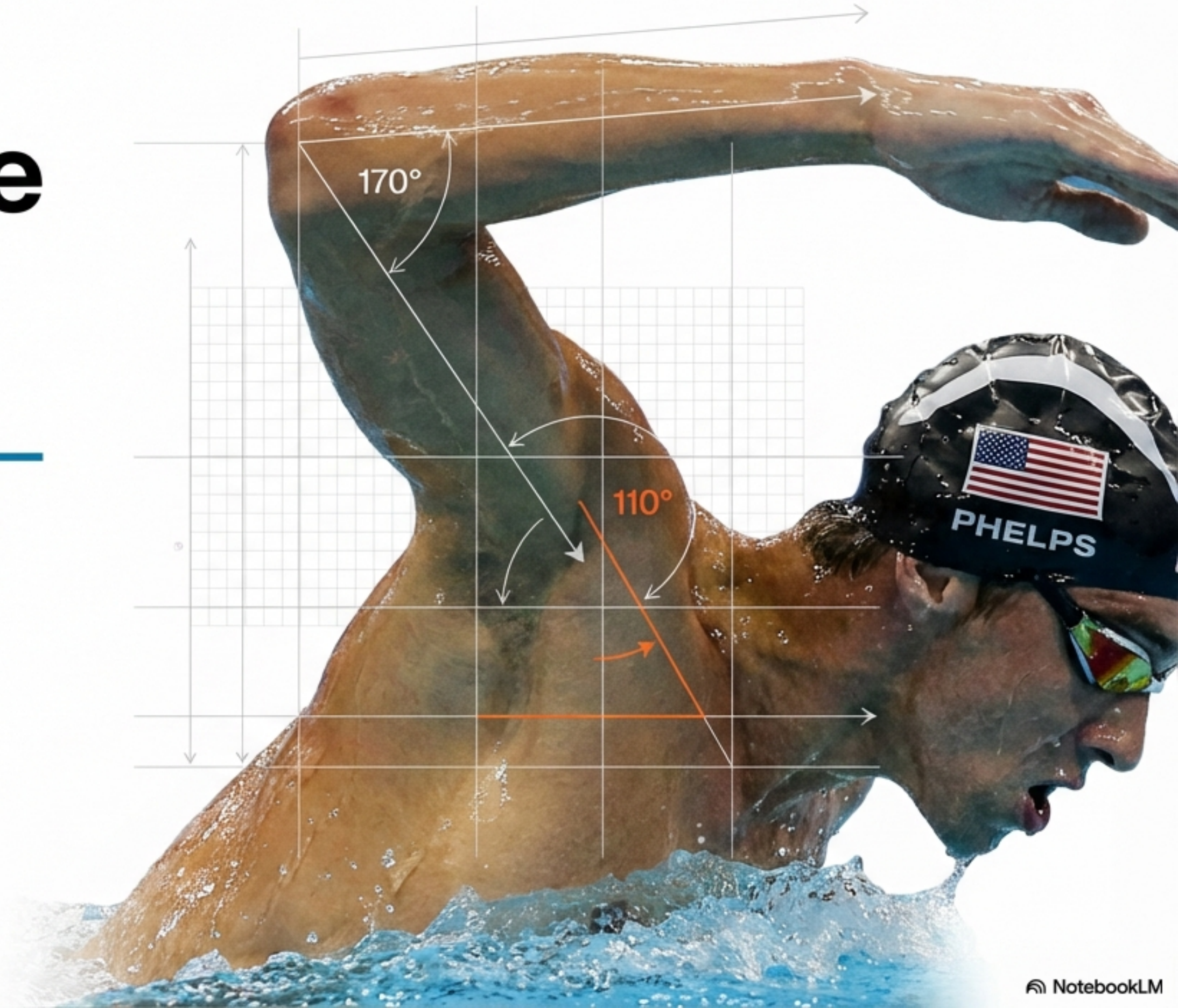


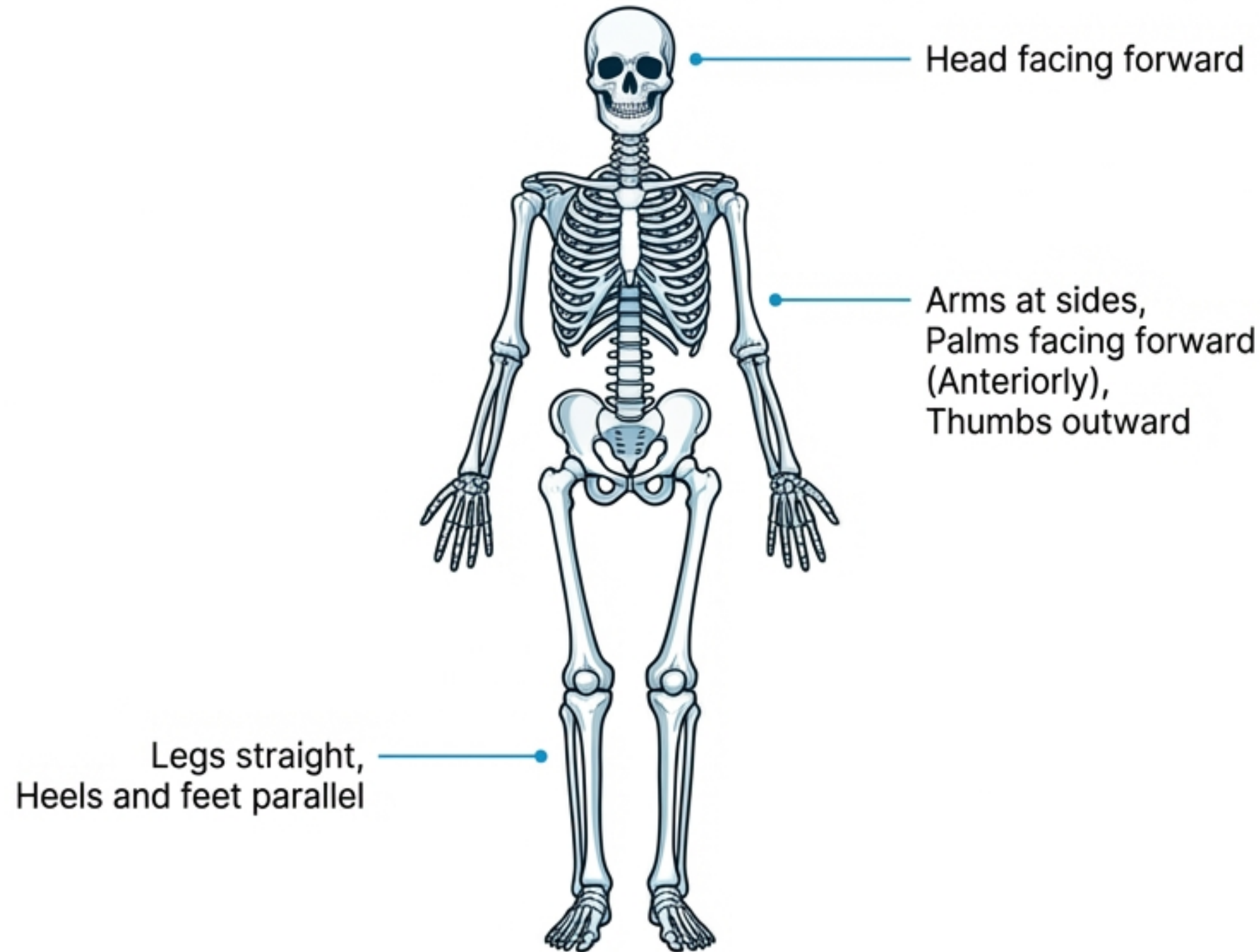
The Language of Anatomy

Foundations of Kinesiology

Kinesiology is the study of human movement. To navigate the geography of the body effectively, we must first learn its universal language—a system of reference points, planes, and directional terms that turns abstract observation into precise scientific description.



Calibration: The Anatomical Position



REMEMBER: This is the “Zero Point”. Always assume the subject is in this position when describing body parts, even if the patient is currently sitting or lying down.

Navigation: Vertical and Depth Coordinates

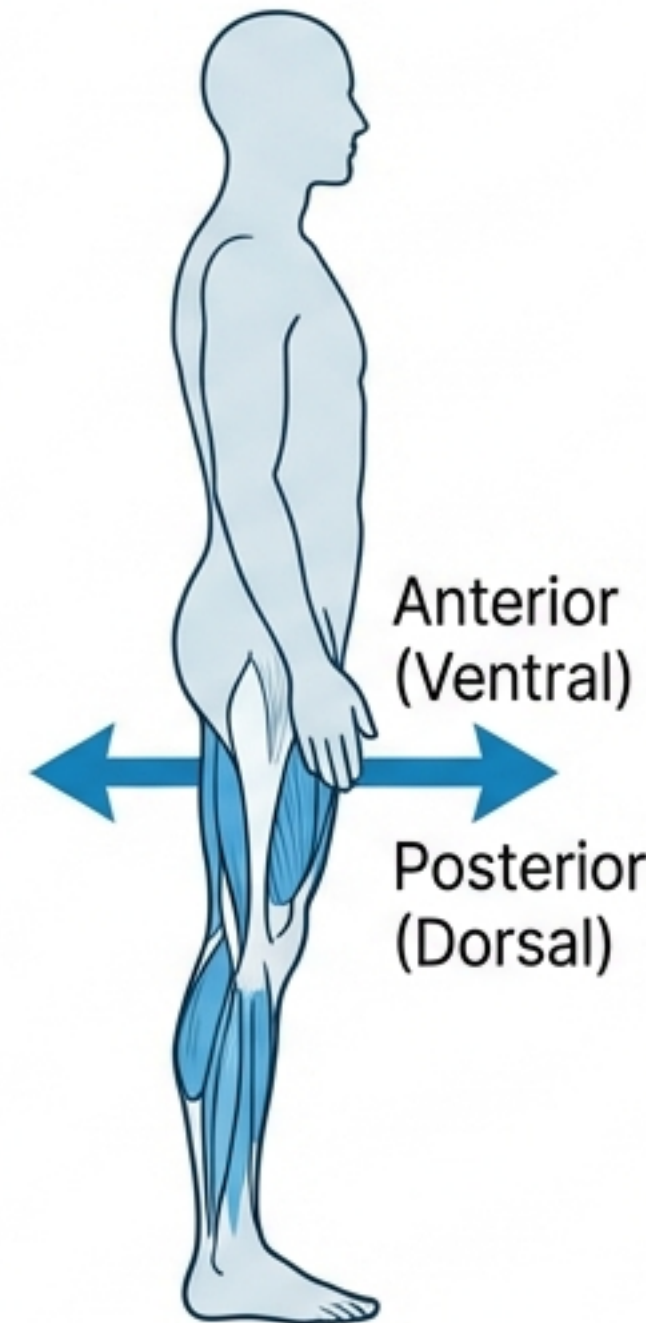
Superior vs. Inferior (The Y-Axis)



Superior: Nearer to the head.
(e.g., The head is **superior** to the thorax)

Inferior: Nearer to the feet.
(e.g., The stomach is **inferior** to the heart)

Anterior vs. Posterior (The Z-Axis)

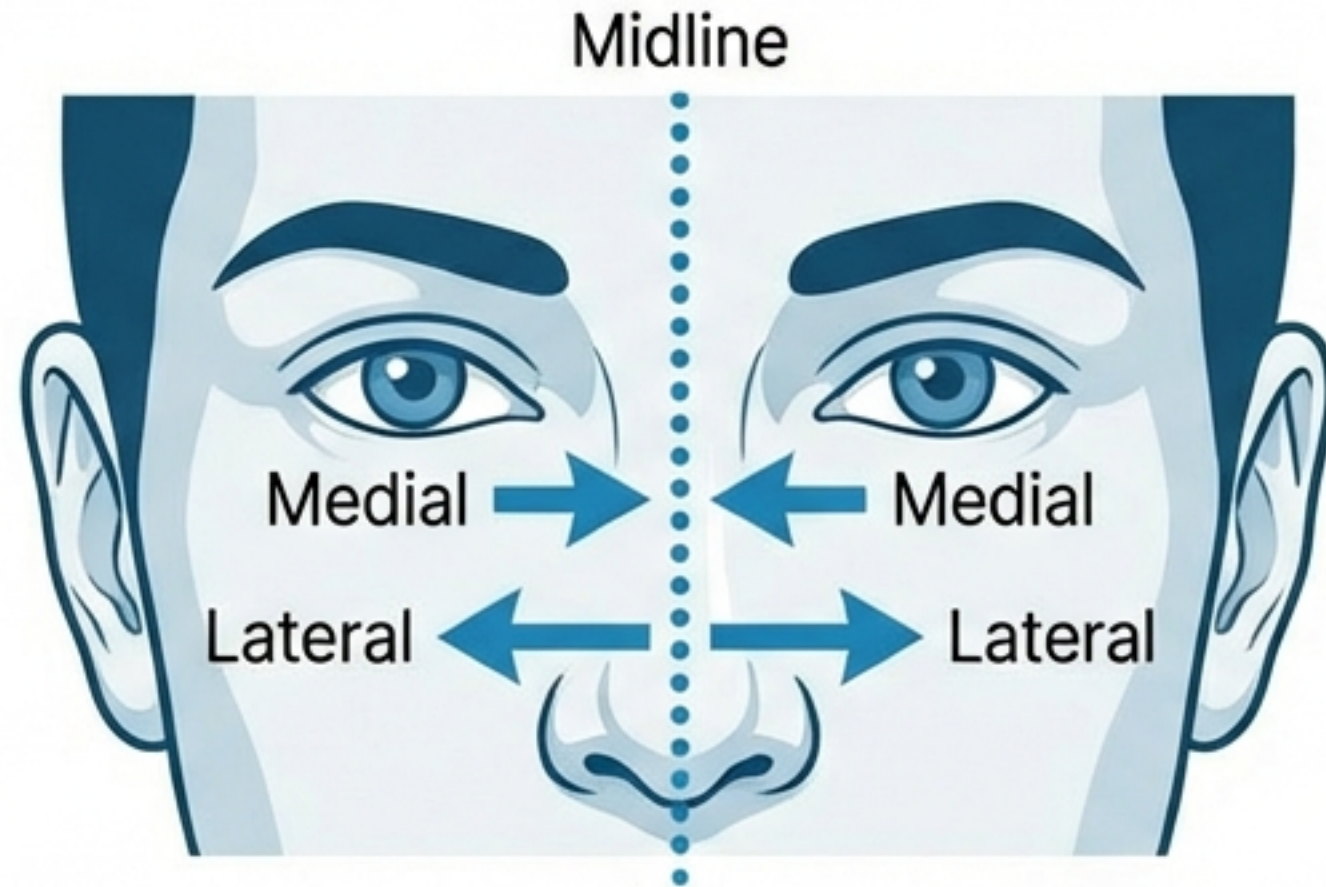


Anterior: Nearer to the front.
(e.g., The **quadriceps** are **anterior** to the hamstrings)

Posterior: Nearer to the back.
(e.g., The **hamstrings** are **posterior** to the quadriceps)

Navigation: Horizontal and Proximity Coordinates

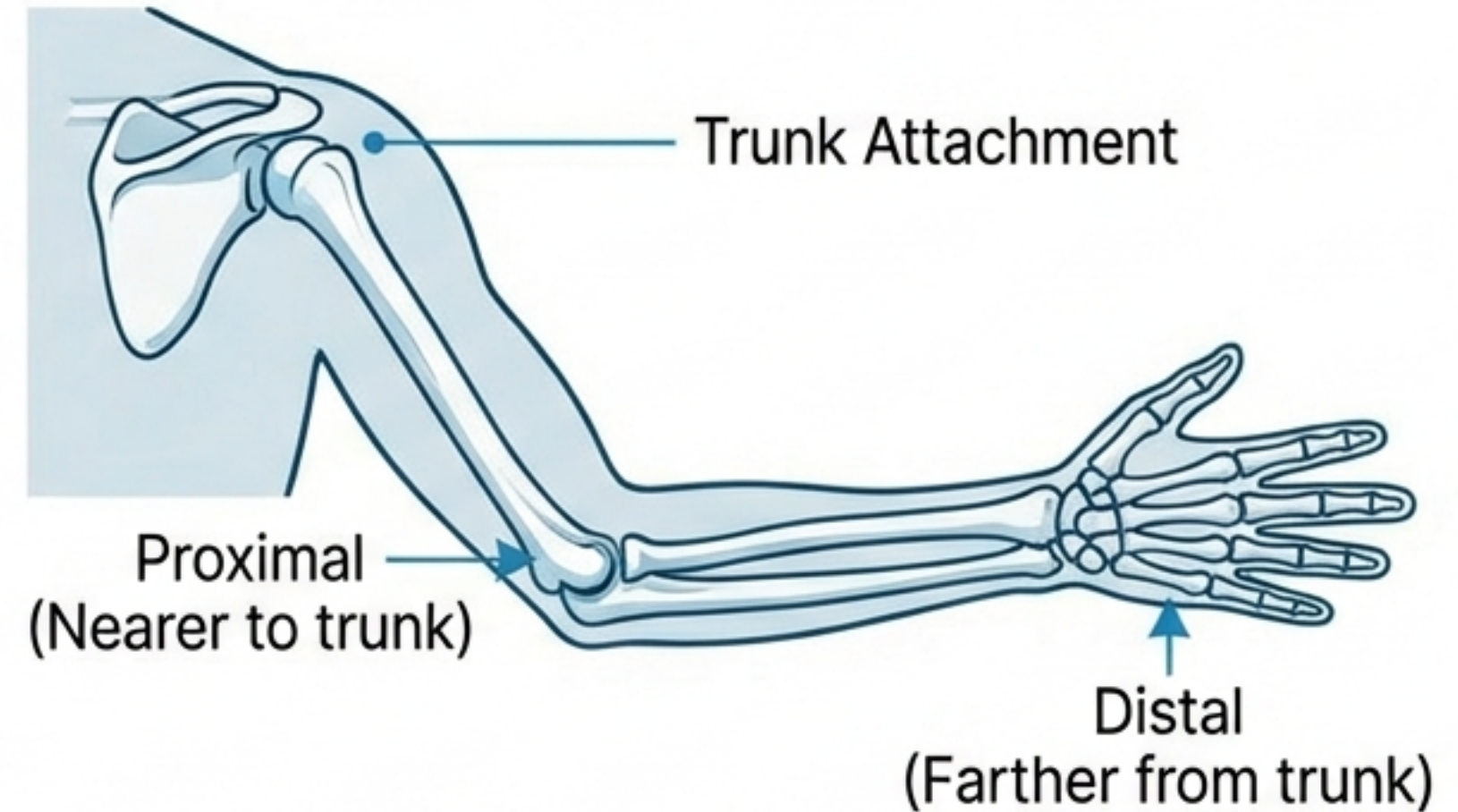
Horizontal Axis



Medial: Nearer to the median plane.
(e.g., The nose is **medial** to the eyes)

Lateral: Farther from the median plane.
(e.g., The eyes are **lateral** to the nose)

Proximity (Limb Specific)



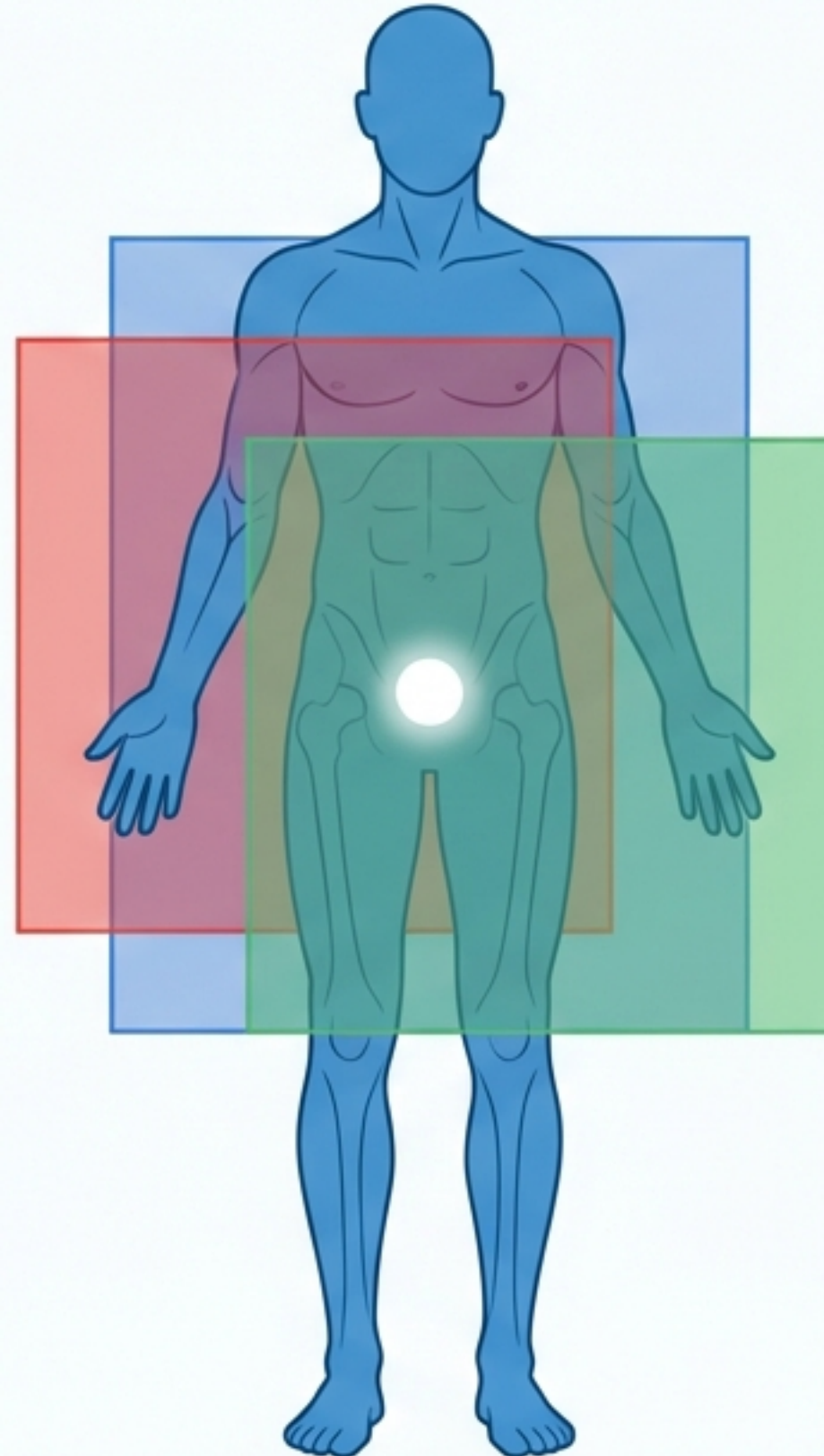
Clarification:

Proximal and **Distal** are used specifically to describe points on a limb relative to its attachment to the torso.

Zoning: The Center of Gravity

The Concept:

The theoretical point where all three cardinal planes (Median, Frontal, Transverse) intersect.

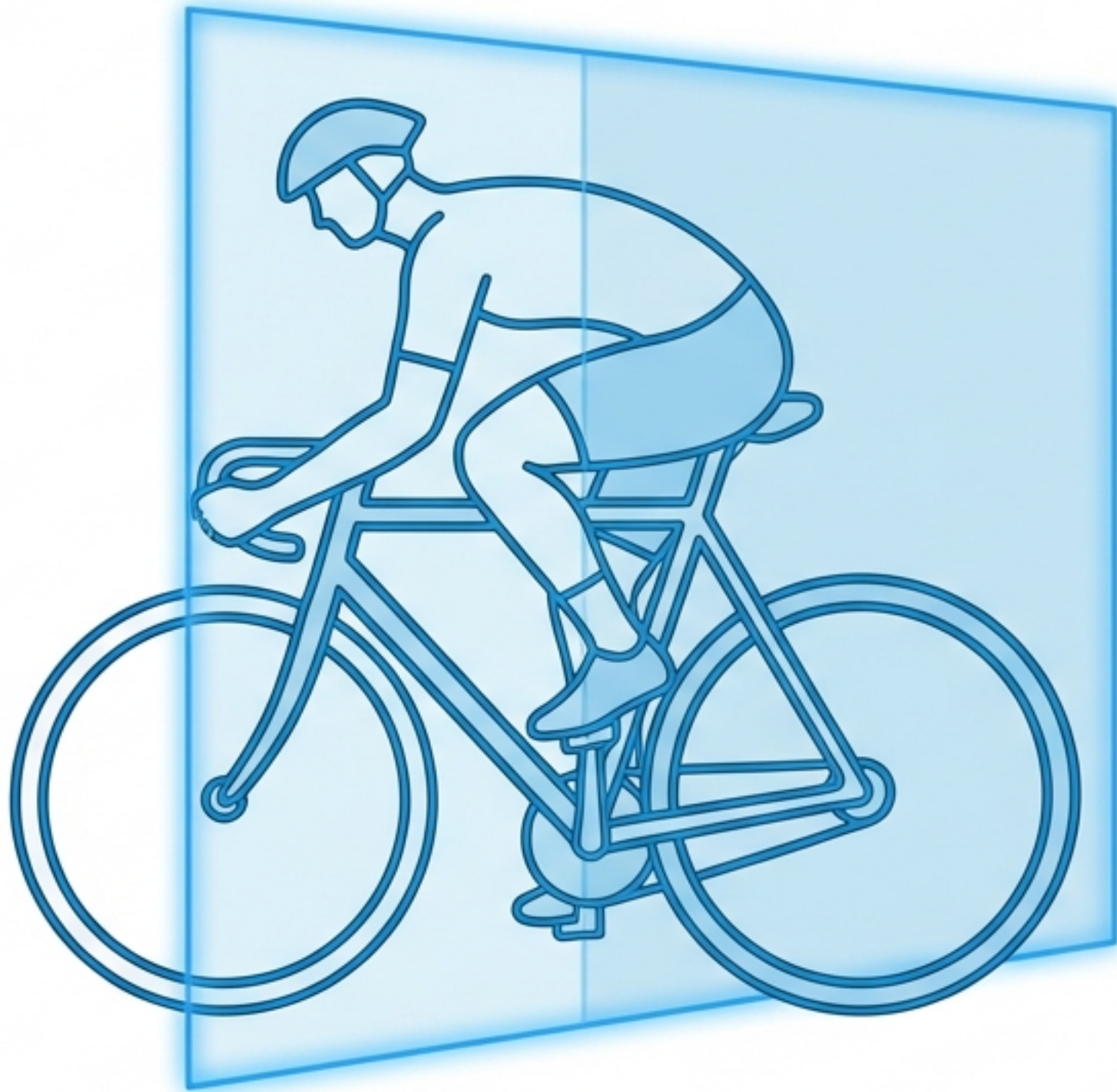


Application:

Represents the center of mass. In the anatomical position, located in the pelvic area, slightly anterior to the sacrum.

Critical for analyzing balance and stability.

The Sagittal Plane: The 'Hallway' View



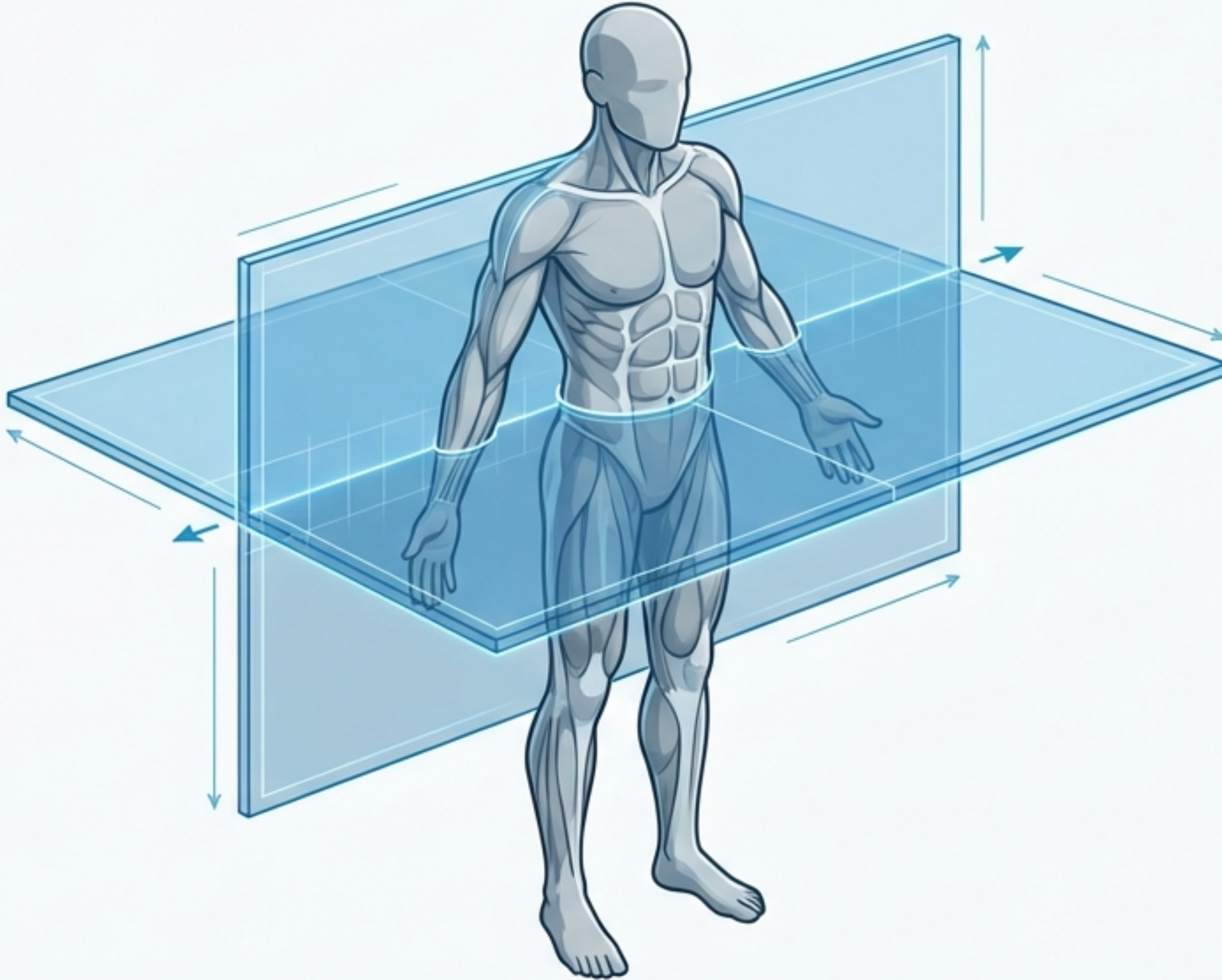
- **The Cut:** Vertical plane running front to back. Divides body into **Right and Left**.
- **Median (Midsagittal) Plane:** Divides body into equal halves.
- **Movement Context:** Forward and Backward motion.
- **Examples:** Running, cycling, nodding, bicep curls.

The Frontal (Coronal) Plane: The 'Wall' View

- **The Cut:** Vertical plane running side to side. Divides body into **Anterior** and **Posterior**.
- **Movement Context:** Side-to-side (Lateral) motion.
- **Key Insight:** Imagine being stuck between two walls; the only movements possible are Frontal Plane movements.
- **Examples:** Jumping jacks, cartwheels, lateral arm raises.



The Transverse (Horizontal) Plane: The 'Table' View



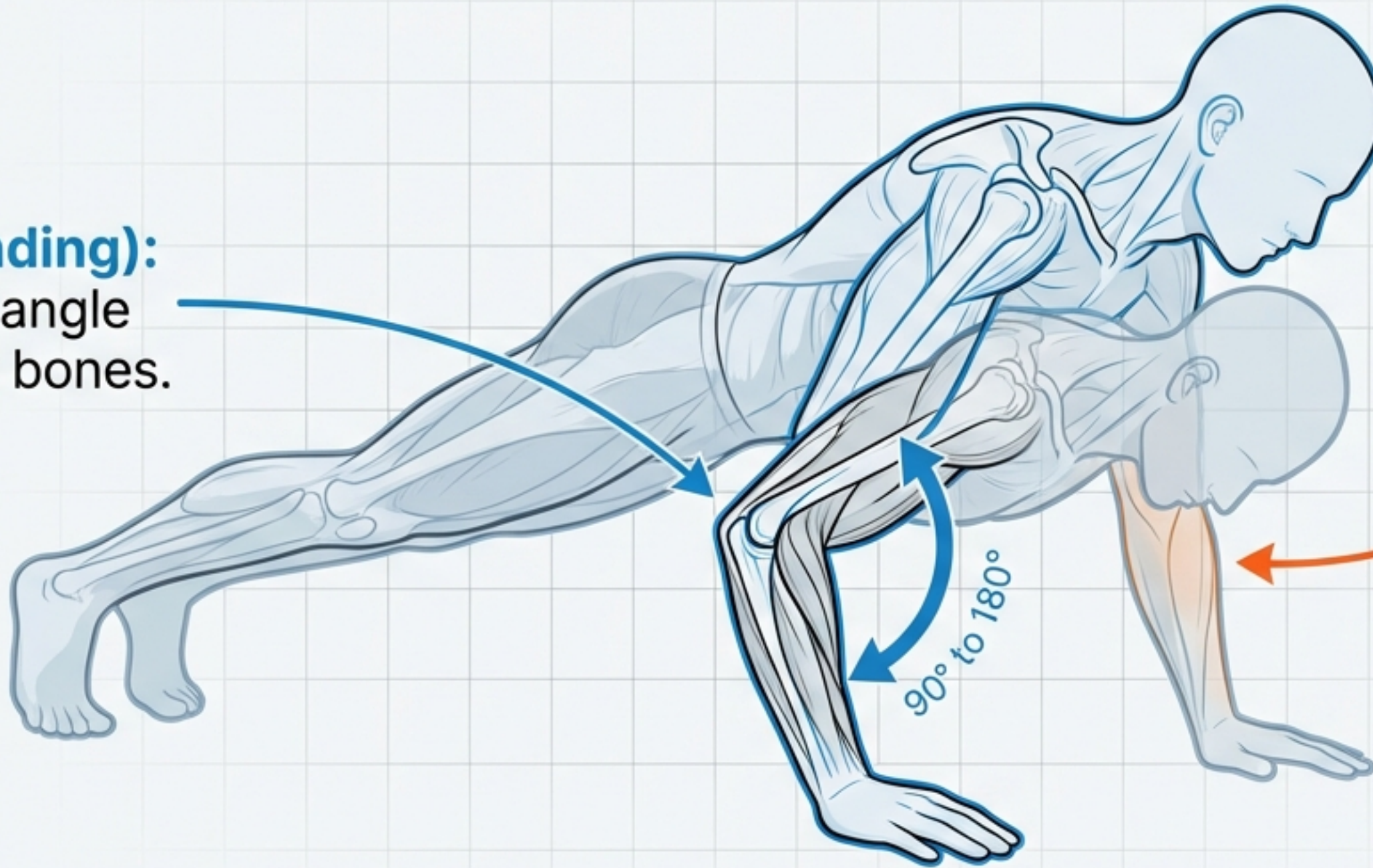
- **The Cut:** Horizontal plane. Divides body into **Superior (upper) and Inferior (lower)** sections.
- **Movement Context:** Rotational motion parallel to the ground.
- **Examples:** Twisting the torso, ballet pirouette, swinging a bat.

Animation: Flexion and Extension

Primarily occurs in the Sagittal Plane.

Flexion (Bending):

Reduces the angle between two bones.

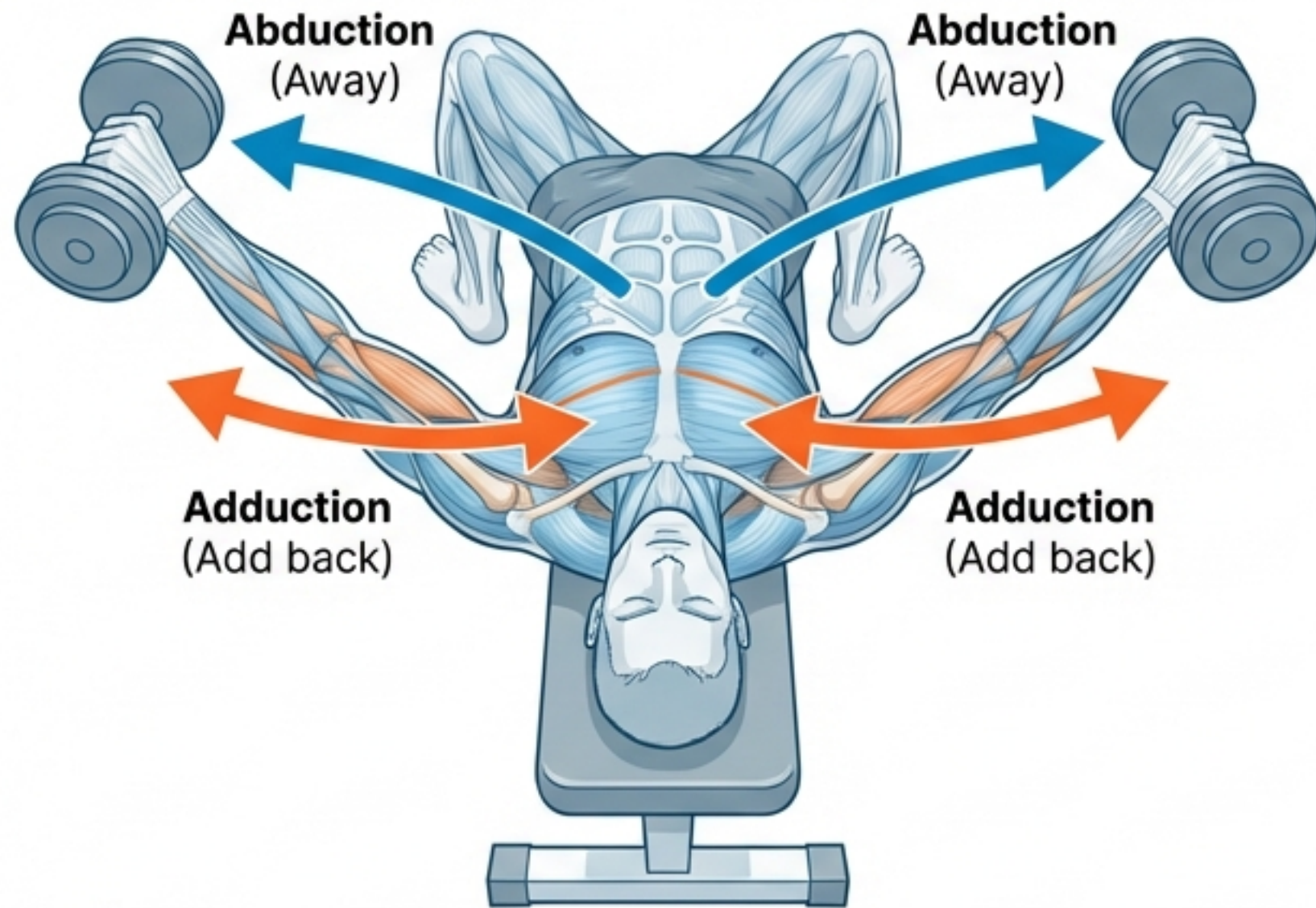


Extension (Straightening):

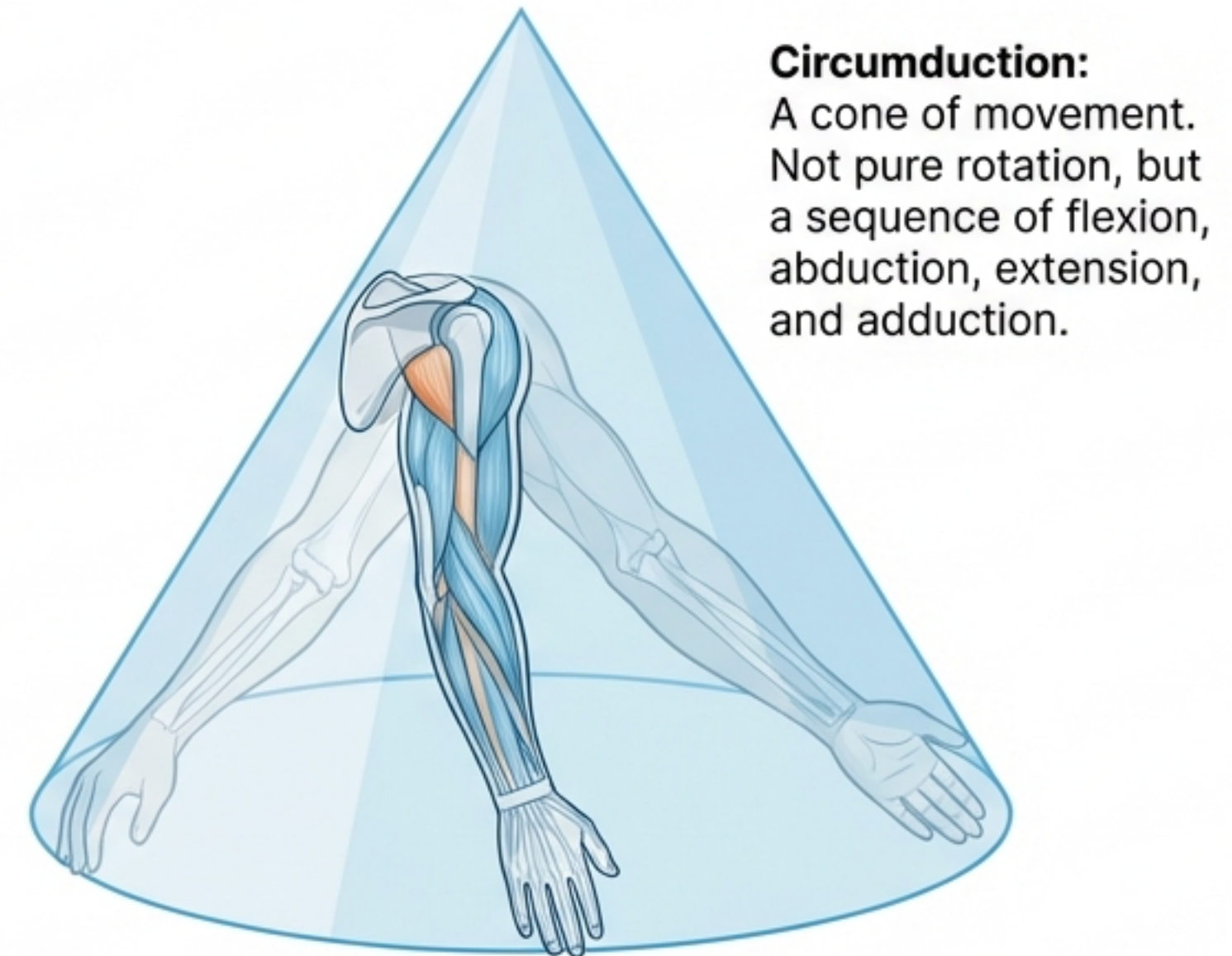
Increases the angle between two bones.

Animation: Moving from the Midline

Abduction/Adduction



Circumduction



Animation: Rotation and Forearm Orientation

Rotation: Turning a bone along its longitudinal axis (Transverse Plane).

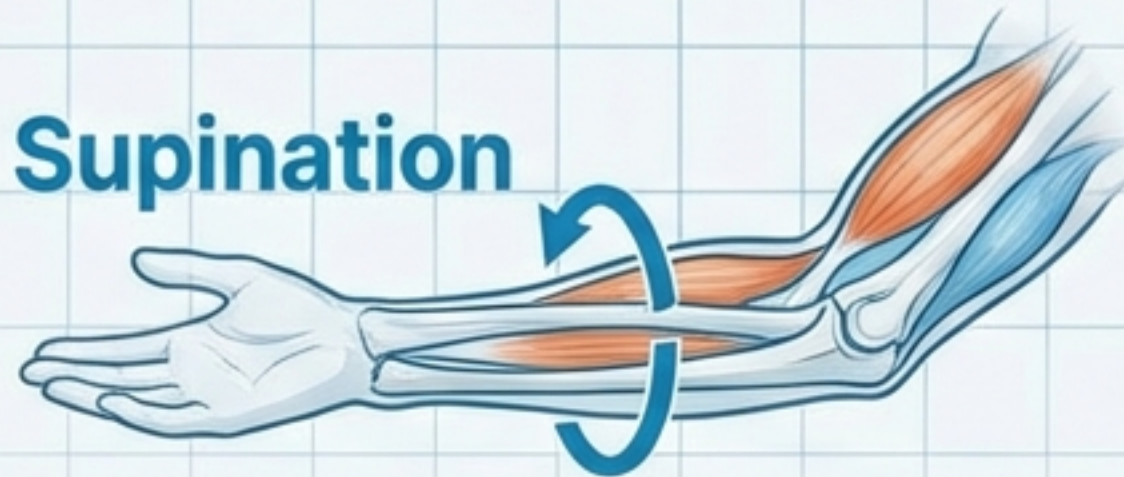


Medial (Internal) Rotation: Turning inward.

Lateral (External) Rotation: Turning outward.

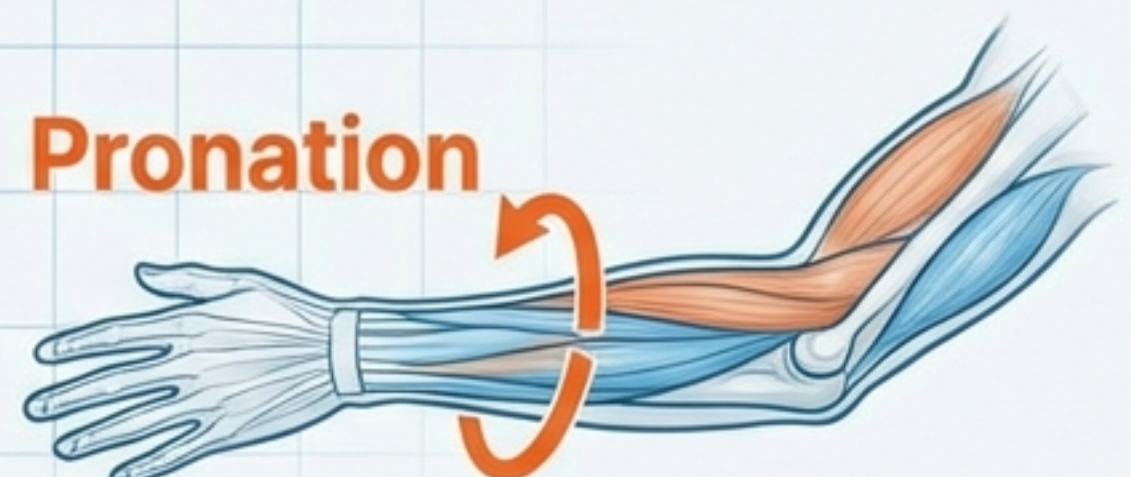
Forearm Special Case

Supination



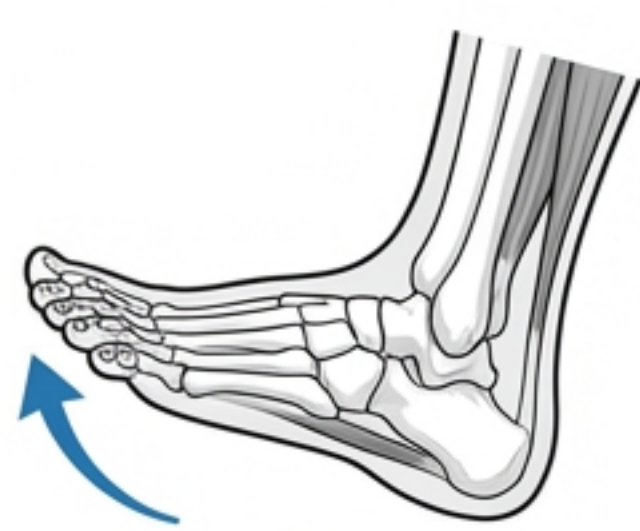
Palm Anterior.
"Holding a bowl of soup".

Pronation



Palm Posterior.

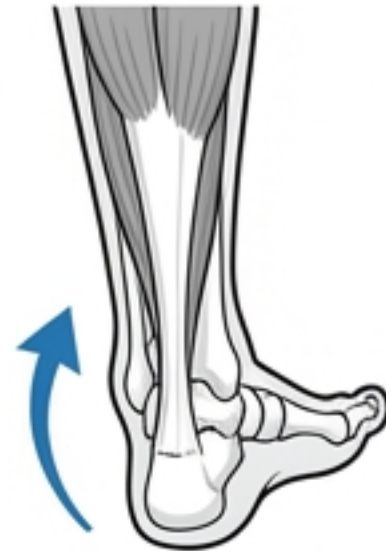
Animation: The Foundation (Ankle & Foot)



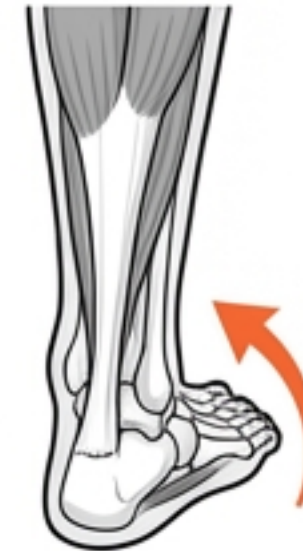
Dorsiflexion
(Toes up toward shin).



Plantar Flexion
(Planting the toes).



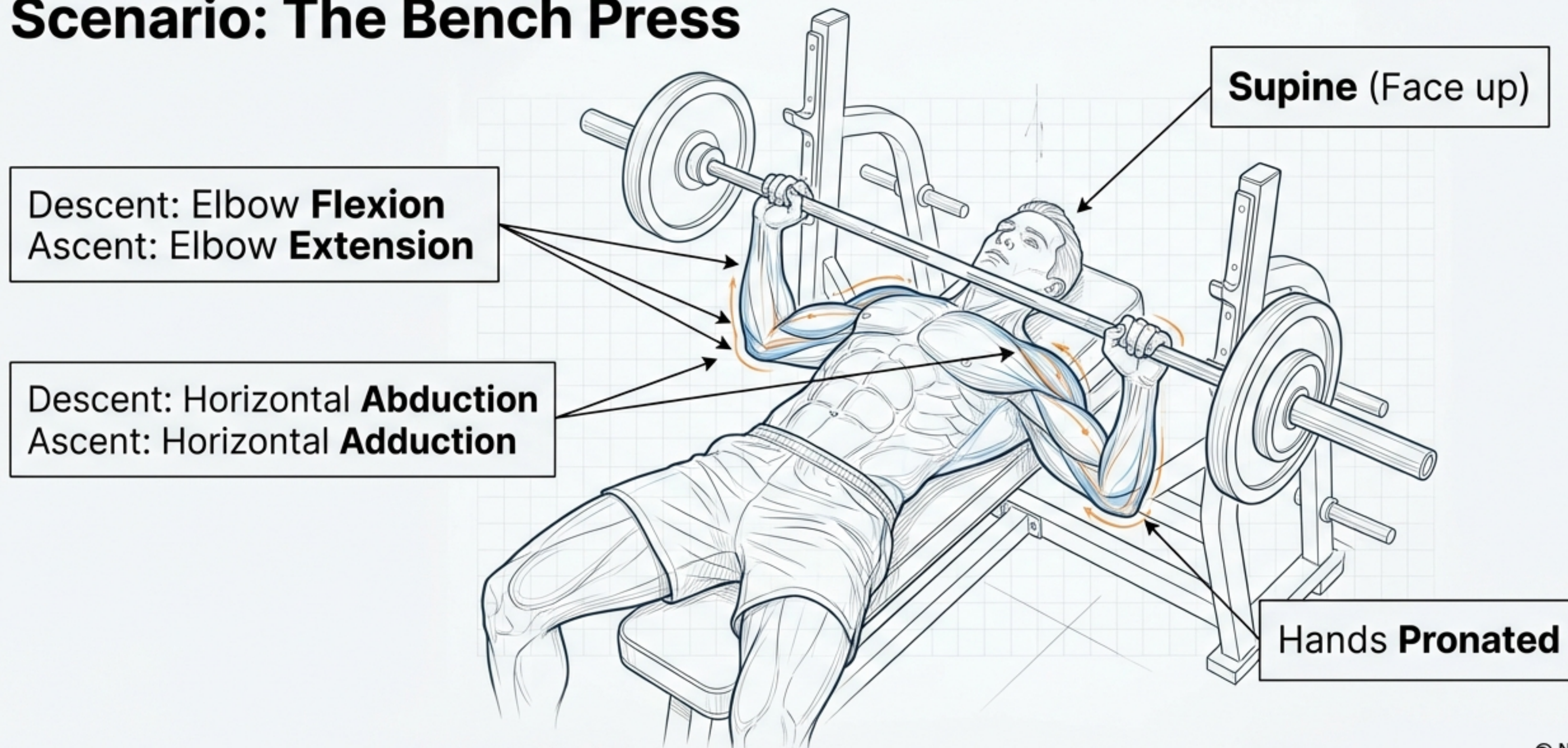
Inversion
(Sole faces medially).



Eversion
(Sole faces laterally).

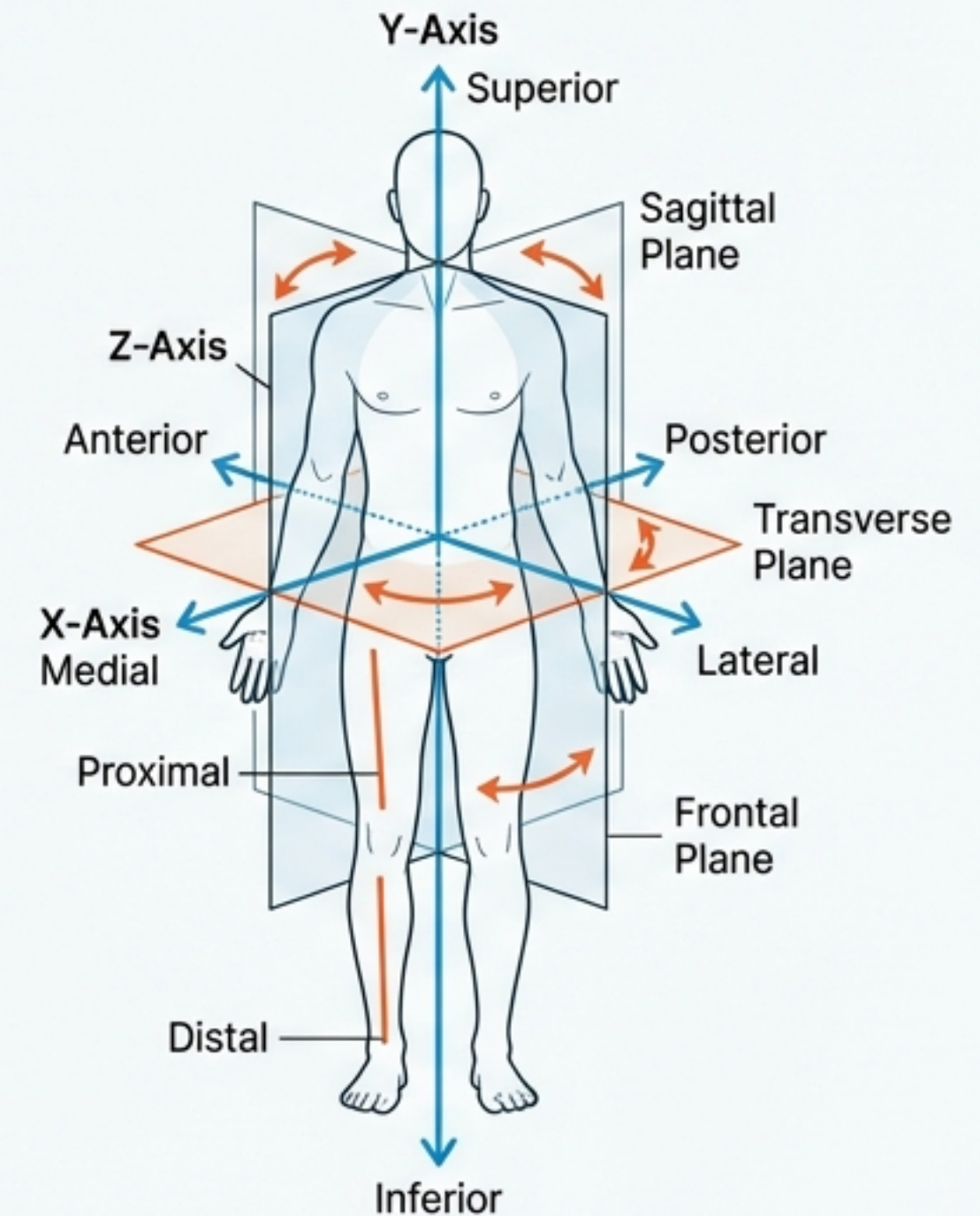
Synthesis: Anatomy in Action

Scenario: The Bench Press



The Kinesiology Reference Grid

Dimension	Directional Pairs	Related Movements
Vertical (Y-Axis)	Superior → Inferior	Elevation / Depression
Depth (Z-Axis)	Anterior → Posterior	Sagittal Plane (Flexion / Extension)
Horizontal (X-Axis)	Medial → Lateral	Frontal Plane (Abduction / Adduction)
Limb Attachment	Proximal → Distal	Transverse Plane (Rotation)





Precision is Performance

Anatomy is not just a list of terms; it is the user manual for the human body. By mastering these planes, axes, and movements, we gain the ability to analyze dysfunction, improve athletic performance, and understand the mechanics of human health.

Structure determines function.